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COS700 Research Report

The Adaption of Perspective Sharing for External Translanguaging within a Smart Learning Environment

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Abstract

As multilingualism becomes more prominent, current Smart Learning Environments do not have the adequate support for cross-lingual communication. This gap has an impact on learning that is inclusive particularly in South African where most students know more than one language and need to strangle in order to participate. The absence of translanguaging tools in smart learning environments stunts students being able to use their linguistic skills. A literature review was conducted to understand the concepts and the knowledge gained was used. Through the principles of Design Science Research, a prototype with real-time speech-to-speech translation and spatial audio [this was to bring perspective sharing to life audible] integrated was built. The prototype was refined by iterative testing in controlled and classroom environments. As testing displayed, there was a somewhat accuracy in translation and audio outputs. There are limitations in current Text-To-Speech libraries for African languages showcased areas for potential development.

Keywords:

Speech-To-Speech Translation, Multilingualism, Translanguaging, Head Related Transfer Function, Perspective Sharing, Smart Learning Environments.

2 Introduction

As the world has become more interlinked, multilingualism has now become the new norm within classrooms. This is reflected in educational environments reflecting the rich linguistic diversity like South Africa . This new multilingual setting offers special opportunities for interactions and cultural enrichment. However, significant challenges do present themselves when trying to make education more inclusive; especially when trying to facilitate effective communication among those who do not share a common language. Communicating effectively remains a vital part of impartial participation, understanding, and engagement within educational settings that are diverse.

Smart Learning Environments (SLEs) have been created to give students a personalised and adaptive experience, thanks to advanced technologies. To ensure that all students are included and to support academic success for said students, SLEs have been crafted to meet their diverse needs. Despite their innovative design, current SLEs lack the apparatus needed to support translanguaging, a scholastic/pedagogical method that allows multilingual students to fully utilise their linguistic repertoire for the improvement of their comprehension and understanding. Not having effective support for translanguaging, the potential cultural and cognitive resources provided by multilingual students remain underused and the inclusive potential that comes along with SLEs is limited.

This research will delve deeper into integrating real-time speech-to-speech translation (S2ST) and perspective sharing within SLEs as an attempt to address these gaps. The tools allow multilingual students to interact with one another in a natural way, creating a space where cross-cultural understanding and engagement can flourish extensively. Additionally, this research will explore how 3D audio technologies can enhance the multilingual experience, allowing students to experience an intuitive and immersive auditory environment regardless of their location.

The subsequent sections of this report will delineate the problem statement, explore the foundational concepts of translanguaging, perspective sharing, smart learning environments and audio technologies and discuss the methodologies used when developing the prototype. It will also discuss the findings with the overall goal of making smart learning environments more culturally responsive and inclusive to better accommodate linguistic diversity found within South African classrooms.

2.1 Problem Statement

Current smart learning environments lack effective tools for translanguaging, a critical function for cross-lingual communication within multilingual classrooms and language acquisition. Despite its academic significance, a gap exists in enabling perspective sharing to support translanguaging within these environments. This project aims to bridge said gap by developing a prototype that integrates Speech-To-Speech Translation and facilitates immersive perspective sharing through the use of Spatial audio technologies, namely Head Related Transfer Function. Through the design, development and evaluation of this prototype, this research will focus on addressing the existing limitations of smart learning tools. A particular emphasis will be placed on enhancing learning outcomes and communication for multilingual students. The main goals are to assess the role that spatial audio technologies play in creating an inclusive, engaging learning experience and investigate the educational goals of integrating Speech-To-Speech Translation. The research's findings will inform the development of accessible and adaptive educational tools that will meet the ever-changing needs of a globalised student body.

2.2 Methodology

This research utilized a dual methodology consisting of prototyping and literature review to effectively address the identified challenges and to fulfill the objectives of integrating perspective sharing for external translanguaging within a prototype. The prototyping methodology was done in accordance to the Design Science Research, a research paradigm conceptualized by Alan Hevner which focuses on artifacts being created in an attempt to solve real-world problems. The literature review offers a comprehensive foundation on current research and theoretical frameworks when it comes to smart learning environments, perspective-sharing, audio effect application, translanguaging and speech-to-speech translation. This underpins the design and development of the prototype, consolidating established research, theoretical perspectives and best practices within the field to support seamless integration in educational settings. The Design Science Research (DSR) approach acts as a framework for creating artifacts to solve real-world problems [2] to make sure that developed solutions are effective and practical. The prototyping methodology will be guided by DSR and will focus on the practical implementation of the designed artifact. This practical approach will allow for constant testing and refinement of the prototype. Doing so will allow for user feedback to be directly addressed and to be able to make necessary adjustments that will lead to having a more effective final product. These

methodologies were chosen based on their complementary strengths. DSR is in line with the objective of creating a solution that centres the user, making it possible for iterative improvements based on the feedback received. The literature review grounds the study in existing research, making sure that the artefact developed is in line with best practices and theoretical frameworks. Prototyping translates the theoretical concepts into a practical application, allowing for effective testing and refinement. Combining these methodologies creates a comprehensive way to solve the issues found within smart learning environments

2.2.1 Literature Review

Literature Selection Process To build a robust theoretical framework for this research, a literature review was done using Google Scholar as the main database. Google Scholar has a comprehensive indexing of academic journals across many disciplines, which helps as this study has an interdisciplinary nature. The following keywords were used to filter relevant papers:

- Speech-To-Speech Translation
- Spatial Audio
- Translanguanation
- Perspective Sharing
- Multilingualism
- Head Related Impulse Response
- Head Related Transfer Function
- 3-Dimensional Audio
- Smart Learning Environment
- Design Science Research

Additionally, Boolean operations (AND and OR) and field-specific filters were done in the refinement of search results. This ensures that the papers selected were the most relevant, peer-reviewed and recent.

Research Areas Covered The literature review looked at different research areas, each important to the objectives of the research:

- **Smart Learning Environment:** Insights into how each learning system has been designed to be adaptive and personalised were gained. Papers on SLEs were important to understand how technology can support learning needs within one platform.
- **Translanguaging:** Papers on translanguaging were looked into to explore the pedagogical strategies that let multilingual students use the repertoire. This idea is a focal point to support the linguistic diversity in SLEs.
- **Perspective sharing:** There has not been much research that delves into perspective sharing. However the current research explained the intricacies of how students can interact in virtual environments whilst experiencing a constant view.
- **Speech-To-Speech Translation:** Papers talking about S2ST were used to gain an understanding of automatic real-time language translation. This gave insight into the limitations technologically and the advancements in trying to support seamless communication across languages.
- **Multilingualism:** Literature on multilingualism was vital in putting the South African educational landscape into context. The complexities and benefits having many languages in classrooms are highlighted
- **Design Science Research:** Literature on DSR was studied to guide the approach to design and evaluate the prototype. It gave a structured framework to iteratively designing, testing and refining the prototype.
- **Head Related Transfer Function(HRTF):** Research into HRTF was selected to explore how auditory cues can be simulated to mimic natural sound localization. This is important in understanding what spatial audio potential could be achieved.

2.2.2 Hevner's Design Process

Problem Identification As stated in the problem statement, current SLEs do not possess the necessary tools to facilitate cross-lingual communication and translanguing. Such absence places a strain in language learning and iterating with people from different cultural backgrounds. Without

adequate voice-based multilingual translation and perspective sharing mechanisms, students and educators could encounter barriers when it comes to communicating, comprehending and engaging with each other. This limitation negatively affects learning outcomes and weakens the inclusivity required in educational settings. The identified gap in SLEs is found where digital inclusivity, educational technology and multilingual communication intersect. As we see SLEs become more and more prevalent, this is a growing need for efficient cross-lingual communication, especially in classrooms where there is a diverse lingual profile. Research within educational technology highlights the fact that inclusive tools are required to support various learners. However, SLEs fall short when it comes to promoting real time speech-to-speech translation and cross-cultural understanding. Research on S2ST and 3D spatial audio technologies suggests that there is some potential when it comes to boosting engagement and comprehension in places where more than one language is spoken. Bridging that gap can be meaningful to inclusive education research by showing how integrating immersive audio technologies and translanguaging can improve learning and inclusivity in educational settings. When prototyping, Hervner’s DSR was chosen as a guiding methodology as it places emphasis in utilising an iterative development cycle, thus always receiving feedback and doing refinements. Given the complexity of integrating spatial audio technology, S2ST and 3D Audio into an SLE, DSR fits perfectly for testing, refining and assessing what impact the artefact has on learning outcomes. Placing focus on problem solving and improvement of the created artefact, this approach makes sure that the prototype is designed and built with practicality and the users’ needs in mind (which makes it adaptive to the dynamic requirements of multilingual educational environments). To guide the prototyping process the following was kept in mind:

- How can S2ST technologies be effectively integrated into SLEs to support multilingualism?
- What role can 3-D spatial audio play in enhancing learning for perspective sharing?
- What educational benefits can be gained from integrating immersive audio technologies in SLEs for multilingual learners?

The project’s primary objective focuses on building a SLE that integrates multilingual S2ST and 3D spatial audio to facilitate immersive cross-lingual

communication between students and educators.

The key objectives are:

- Creating an interface that is user-centred and intuitive. This will allow students to easily use the program and allow for translations to be experienced using perspective sharing.
- Enhancing the accuracy of translation. This will make sure that there are high-quality, accurate translations across multiple languages (especially South African languages). This allows for clear, contextual translations which is important when in an academic setting.
- Ensuring effective speech synthesis. Having a clear audio output in mind and attempting to come up with effective and natural sound speech synthesis, this plans to improve the student's comprehension and allow them to more easily engage with the content that has been translated.

Having the user in the forefront when in come design the prototype shaped the project's goals to ensure relevance and easy use. The needs of the user consist of having an intuitive interface, having real-time translations that are reliable and support for languages that can be found in different classrooms [for the purpose of this study, we will focus on isiZulu and Afrikaans as our 2 languages]. Addressing this, the accuracy and clarity of the quality of the audio and language output was the main emphasis.

However, multiple technical constraints have influenced the way the prototype has been designed and developed. For instance, the limited text-to-speech (TTS) support for certain South African languages brought issues when creating accurate audio translation, In addition implementing 3D spatial audio needed significant processing power and compatibility needed to be taken into account especially when being used in real time. These requirements showed the need for having flexible design solutions and helped in the selection of specific tools and libraries. There were several practical constraints that shaped the project's objectives such as library limitations, the compatibility of different technologies and budget constraints. Some S2ST and TTS libraries did not have the adequate support for some South African languages thus restricting options for seamless translation. There were compatibility issues as there were differing Python versions needed for certain speech synthesis tools. Monetary constraints limited access to commercial TTS services, thus needing to use open-source alternatives with varied functionality. These limitations were noted when formulating the objectives making sure the goals set out were realistic and achievable while trying to achieve a high-quality artifact which is user centered.

Artefact Development The artifact was designed using a design approach that is iterative, which allowed for constant refinement and enhancement based on the feedback received and testing performed. Initially, the development began with each mini prototype of basic speech-to-speech translation and basic audio processing being built. Once they worked sufficiently, they were combined into a large prototype. Over multiple cycles, the prototype was fine tuned to improve translation accuracy (such as testing out phonetics)etc, incorporating 3D audio for an immersive experience and adding support for more languages. Each iteration looked to have a sum of improvements. Early versions focused on refining the quality of audio output while later versions made sure that the user interface was responsive and that languages were translated accurately. This approach was important as integrating spatial audio and S2ST has its complexities as every new feature added brought in new challenges that required testing and modification.

Feedback was collected throughout the development process from testing sessions and review*. Initial feedback showcased that translation accuracy especially with isiZulu could be improved. Adjustments were done to refine the translation algorithm, improve clarity and reduce latency in response times. There were specific adjustments in accordance to interface elements to make them more intuitive for users. This includes showing the written translation (in the event students can not hear their volume) and adjusting volume when using spatial audio effects. In addition translation output was attempted to be refined to improve the accuracy of complex sentences which can be useful within the classroom. These changes were done to address the usability of the artifact and to improve student engagement and satisfaction. There were multiple challenges when developing the artifact.

Technical limitations of TTS libraries were a notable speed bump, especially within limited support for African languages. Integrating certain TTS and audio processing libraries such as Mozilla TTS raised issues when it came to compatibility. This was because each library needed a certain version of Python and could not adapt to the latest version of python. These challenges emphasised the need to be flexible when in development as it was a necessity to make constant switch to different libraries and tweak the artefact's technical requirements. Despite this, each iteration provided insight that allowed for gradual adjustment making sure that the final artefact was not only user friendly, but functional. The developmental process was precisely documented and tested, with every iteration having detailed testing and evaluation. Every mini prototype developed was subjected to assess-

ments testing how accurate and how well it performed. Testing looked at how accurate every translation was, what was the quality of audio and how could one experience this artefact (in a classroom setting). This strict approach made sure that each part of the artefact was extensively tested and improved upon when needed. Documenting every change and outcome when testing made it possible to have a high level of transparency and adaptability, ensuring that when new issues arose, they were dealt with in the correct manner.

The artifact was evaluated through a series of testing protocols that were created to validate its effectiveness and performance. Language accuracy tests were conducted to measure how correct each translation was across supported languages. A set of sentences that could potentially be used within a classroom were used to see how accurate their translations were. Audio quality checks assess how consistent and clear was the speech output, particularly with spatial audio effects in place. Testing was done in both controlled and real-world environments. The controlled environments such quiet rooms [isolated rooms within a student accommodation] allowed for audio checks to be prioritised while testing in a classroom environment [a small to medium sized classroom with 15 people] helped evaluating the artefact’s performance in a typical classroom. The success of the artefact was measured against these metrics:

- How accurate translation was: This is the percentage of correct translations on actual references, with a target of 80% or high being the target accuracy
- Response Time: this is the measure delay in audio input and translated output, aiming for under 3 seconds when being used in real time.

During testing and development, phonetics were introduced as a method to improve how translate isiZulu sentences were articulated in an attempt to bridge the gap between the limited support of African languages. While simple sentences such as “Good day, my name is Joy” achieved an accuracy of 70 percentage (nuances such as time of day and who is being spoken to would need to be considered), clarity of phonetics was decreased as sentences got a bit more complex. This was largely due to the difference in sounds[3] between Latin-Based languages [which is primarily supported when working with language libraries] and Bantu languages. Additionally library compatibility issues arose, as certain libraries needed a specific version of Python.

Switching between versions, at times, brought in new errors that clashed with other dependencies. Repeated adjustments were needed to solve these issues within the development environment and libraries had to be vetted so that there is a balance between stable functionality and language support.

An important component of the development process is documentation. This is used to track how much progress has been, reflect on the design process and record the outcome of testing. Throughout the iterative process, doing documentation ensured that improvements and every change done was logged comprehensively, showing what decisions were made and the rationale behind them. This report serves as a vital player in communicating what has been found showcasing the successes and limitations of each design iteration. This will make the artefact be well documented and ready to be practically implemented and further refined if needed. This also allowed for feedback reviews and testing to be integrated which is essential when making adjustments to the artefact. Specific feedback showed areas where audio be more clarified, translations to be more accurate or where interface elements can be intuitive.

Prototyping

Prototyping was an integral part of development allowing for small but eventual adjustments and improvements based on regular testing and feedback. Every small prototype was a basis to enhance and refine parts of the artefact. This was essential to manage the project complexity and every stage allowed for improvements to be targeted. Every prototype iteration brought in new features and improves with specific technical adjustments and design adjustments such as:

- Early iterations started with fine tuning language algorithms for translation accuracy, then optimised audio processing
- User interfaces were gradually refined with layout adjustments made and simplified navigation to allow for easy use.

Enhancements to improve user experience were important to accessibility and engagement. This made sure that it was an intuitive, immersive experience that effectively supported cross-lingual learning.

3 Perspective Sharing

Perspective Sharing, which is still in the early stages of exploration, is a fairly new concept that possesses potential for enhancing engagement in learning environments. As conceptualized by Pieterse and Marshall , perspective sharing allows for multiple users to view, explore and interact in a shared 3-Dimensional virtual environment. This feature was created as an attempt to bridge geographical gaps. This is powerful within educational settings as users in different locations would be able to engage in learning activities that are collaborative and interactive. As students will be placed into the same virtual environment, Perspective Sharing is able to foster a presence and connection. This will allow for richer, shared experiences in training and learning contexts.

Pieterse and Marshall [4] go on to discuss the development of a functional Perspective Sharing system. For a successful implementation, a range of advanced technological components would need to be integrated. Through rapid prototyping and exploratory research, they were able to create a proof-of-concept model for virtual environments with elements that have been specifically identified and tested. The aim was to design a system that was flexible and was able to be applied across diverse fields . This would make sure that it was customly made for different learning needs. By doing so, the platform would be able to be used in many disciplines.

A key feature of Perspective Sharing is the interaction between the virtual environment and its users. Within their study [4], to enhance immersiveness, objects were programmed to follow real-life physical laws in the scene. The users of the system should have the ability to engage with the environment consistently. This will allow for users to have real time reactions to the actions and movements within the system. The dynamic and responsive design aids users in feeling more integrated within the scene.

Within Perspective Sharing, all participants are usually in remote locations but they are able share a synchronised experience of a virtual space. Every user's local display always receives data streams that updates the 3-Dimensional environmet's rendering. This will ensure that all users have a real-time and unified view. This feature lessens discrepancies between users' views, making it possible for users to interact within a consistent version of the shared space, despite of where they are located.

There are many configurable viewpoints which are each suited for different collaboration and learning goals:

1. Round table Perspective: This is suitable for brainstorming. This setup puts users as if they are seated at a table, Allowing for ideas and visual input to be shared freely.
2. Conjoined Ego Perspective: This was created for guided learning. This allows for teachers to lead students through a scene giving a synchronised shared perspective.
3. Roaming Scout Perspective: This is created for the purpose of exploration. Users will be allowed to navigate the virtual environment to look at different aspects of the scene.

To further enhance the immersive experience of perspective sharing, an audio-based approach to perspective sharing can be achieved through the use of Head Related Transfer Function(HRTF).

3.1 Head Related Transfer Function

According to Olman [32], Head Related Transfer Function (HRTF) is a key auditory process that allows for people to perceive where different sound sources are located, especially when it comes to elevation. To understand how this process in detail, the following is the step by step mechanics of how HRTF allows for the sound's origin to be pinpointed by the brain[36]:

1. Sound diffraction and head shadowing is important in perceiving sound. As a sound wave travels towards the listener, some parts may be obstructed by the head. This effect is more pronounced at higher frequencies. This is because a short wavelength leads to a changed and quieter signal opposite the sound source. This enhances sound localization and spatial awareness.
2. The outer ear -pinnae- shapes sound waves through diffraction and reflection. The unique folds and contours change the frequency response based on the angle at which the sound arrives. This leads to changes in phases. Such modifications are important at localising sound specifically on the vertical plane.
3. HRTF does the processing of binaural cues which are Interaural Time Difference (ITD) and Interaural Level Difference (ILD). Essentially,

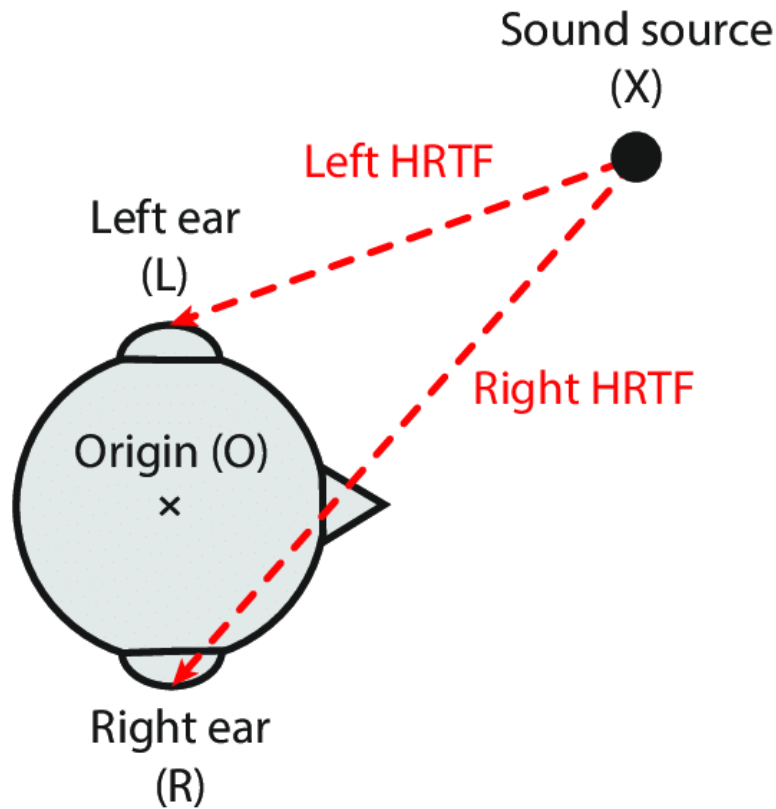


Figure 1: Concept of HRTF visualized, adapted from [1]

ITD shows the delay in each ear getting the sound whilst ILD looks at the difference in sound levels. Combining them allows for the auditory system to determine the horizontal location of sounds.

4. Every listener's anatomy determines how different frequencies are perceived. With HRTF, the variations are mapped usually 20Hz to 20kHz. This makes each sound direction unique.
5. HRTFs are measured using the microphones in the mannequin's or subject's ear canals. They are positioned in environments that are acoustically controlled where sounds are received in different angles. Data gathered from gets the directional response of the auditory sys-

tem, making a detailed map of how sound is filtered based on where the source comes from.

6. As people are different anatomically, HRTFs can vary from person to person. General models used the averaged data from standardised HRTF databases.

HRTF within smart learning environments could help in better enhancing how students learn. HRTFs can replicate 3D spatial audio effects that reflect how natural sound can reach a student's ears from different places and distances. It enables students to experience the spatial parts of lessons where they would need to be together. But before getting there, let's explore the concept of Smart Learning Environments.

3.2 Smart Learning Environments

Smart Learning Environments (SLE) as defined by Hwang [10], are tech-enhanced educational settings that leverage wireless, mobile and context-aware technologies for adaptive, personalised learning.. They make sure that relevant and timely support by receiving the student's real-world status and environmental status. The features that define what SLE are as following[10],[11],[22],[25]:

- They are context aware. Meaning that they are able to traction the real-time positions of students and are able to quickly adapt to the different context scenarios.
- They can perceive the social relationships students have with each other.
- They can predicate what students may need and offer access and easy access to resources and services.
- They allow students to have a fully immersive, dynamic and technology-infused experience
- They ensure that services are uninterrupted across various connected devices.
- They assist with intuitive and multi modal interactions.

- They can develop standards that will make sure that there is seamless integration across various devices, services and platforms.

Within SLEs, students can gain unique advantages such as:

- Students would be able to receive immediate feedback and support through data being analyzed and collected in real time. This makes sure that they are able to stay on tack and adjust their learning strategies for academic success[25].
- Student engagement and learning outcomes are improved as they have personalised and adaptive learning [25][22].
- As multi sensory and interactive technologies are being used, learning is made to be more enjoyable and engaging for students[24][28]. This conjures a positive attitude towards learning.
- Students will be able to learn anywhere, anytime thanks to SLEs.[23][29] This allows for students to learn on their own schedule and makes learning more accessible.

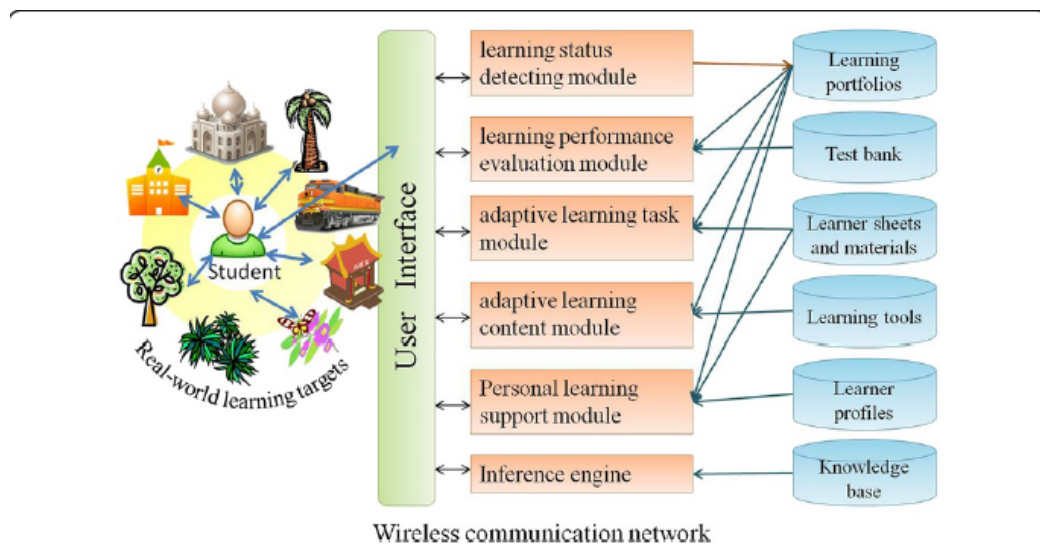


Figure 1 Framework of a smart learning environment.

Figure 2: Hwang's Framework of an SLE[10]

With every benefit that comes with SLE does follow a list of limitations:

- If there is not an equal distribution of SLEs to students, there is an alarming risk of inequality[21][24] especially for those from a disadvantaged background.
- For SLEs to be effectively utilised, teachers would need to be proficient with the pedagogical approaches and technologies that are involved. This will constitute having educators to receive additional support and training[31].
- The advanced technologies for SLEs that will need to be put in place and maintained can be costly and complex[30][25]. This could be a barrier for educational institutions that currently do not possess the monetary resources to do so.
- The safety and privacy of students' information may be a concern [23][29] as devices are interconnected and data is extensively used.

As the foundational elements of SLEs and the transformative potential for education, it is vital to explore how translanguaging fits in and benefits using it in these environments

4 Translanguaging

As defined by Garcia and Wei [4], translanguaging is a dynamic system where people who know one or more languages smoothly use their linguistic repertoire in an educational context. Compared to traditional bilingualism, translanguaging looks at many languages as one large system of languages that students draw from to improve their cognitive flexibility, cultivate identity and expand on their comprehension. This differs from code switching, which is defined as a person switching from a linguistic code [this could be a dialect (a way of speech found within a region) or a language] to another [16]. An example of code-switching inspired by [14] would be, in a classroom in Kwa-Mashu, Kwa-Zulu Natal students usually switch from the language of medium which is typically English to their native language of isiZulu. This would be speaking to their friends in isiZulu and when interacting with faculty they would speak in English.

The social cognitive and cognitive benefits reaped from translanguaging are accentuated by the language pedagogy model curated by Cummins [2]. This links being able to translanguaging to improving inclusivity and being

able to adapt. Cummins [2] advises that students are able to navigate and bargain meaning across languages thanks to translanguaging. The study done by Lewis Jones and Baker [15] supports this, calling attention to how students are able to understand complex concepts using translanguaging, bridging gaps between languages and culture. The study also shows that retention and comprehension are augmented specially when educators actively support this whilst teaching. This is highly beneficial within South African classrooms, where it is common for students to switch from English to their native language and vice versa.

Translanguaging goes beyond education, allowing for cultural fluency in different societies where many languages are spoken. As noted by Gracia and Wei[4], translanguaging helps people work their way through social spaces which impacts societal norms around using language and inclusion. In multilingual countries like South Africa, translanguaging backs up social cohesion by allowing students to smoothly interact regardless of linguistic boundaries. This adaptability is vital as globalization grows the need to be agile linguistically and culturally.

4.1 Speech-To-Speech Translation

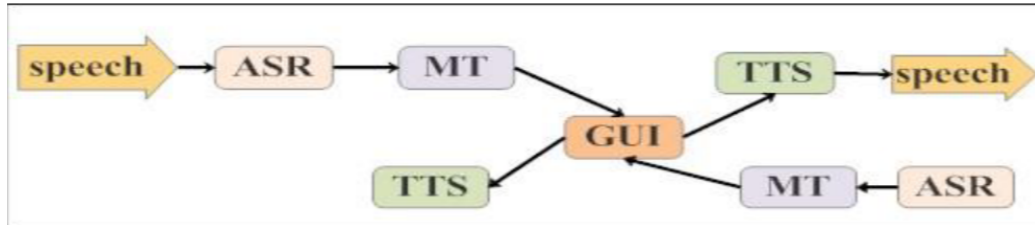


Figure 3: Diagram of how Speech-To-Speech Translation is done [15]

Speech-To-Speech Translation as discussed to by Dhwan[15], is the process of taking a spoken language and converting it directly into another spoken language. This process allows for real-time or approximate real-time translation to happen.

This involves many key steps [37]:

1. Automatic speech recognition technology will capture the words spoken by the user and change them into text. This needs to be highly accuracy as people have different accents, dialects and expressions.

2. The transcribed text is worked on by a machine translation system - usually done by neural networks- and the text will be translated into the require language. This needs the system to accurately interpret the context, syntax and grammar to correctly convey the intended meaning.
3. Finally the translated text will changed into speech through the use od Text-To-Speech synthesis. This makes a spoken output in the target language whilst maintaining the intonation and prosody of the initial input

S2ST has the potentials to break down barriers set by languages in different contexts. It being able to translate in real time allows for smoother and inclusive interactions which can be beneficial in a classroom situation. Without having the need to be fluent in many languages, S2ST aids in the creation of equal access to information across language divides.

However Dhwan[15] does say that S2ST faces issues when trying to attempt full accuracy. Specifically translating idiomatic expressions,slang or references that need to be understood culturally.Also, having different accents, background noise and varying speech speed can hider how accurate recognition can be. A substantial amount of processing power is needed when attempting real time translations staring mobile devices and may need robust internet connectivity. Culturally context and context tends to be lost which is an issue especially when it is need in classrooms in South Africa. Finally, the ethical and privacy implication of the recording processing and keeping spoken data remains as a high concern. This is as depending of where this technology is used, there might be sensitive information being dealt with

Now, with an understanding of these concepts, combining them into a prototype will be followed into the next section.

5 Application of Model

The prototype is named Classify Speech, inspired by the idea being used within the classroom. It currently works in 3 languages, English, isiZulu and Afrikaans.

5.0.1 Link To Prototype:

<https://github.com/jazbengu/ClassifySpeech>

5.0.2 What was used in development

The development and testing of the prototype happened using a 2021 Huawei D14 Notebook. This device operates on a Windows 11 operating system and features an Intel Core i5 processor. It is equipped with 512GB SSD and 16GB of RAM. This allows for a balanced performance whilst working on the move. The solid state drive played a role in fast data retrieval and storage operations. This not only reduced latency when processing data but it allowed for smoother transitions when doing audio manipulation, prototype experiments and evaluations.

The coding implementation was through the use of Visual Studio code. This IDE is lightweight whilst also being versatile. It was chosen for its flexibility and extensive library of extensions that were utilised through development. It offers integrated Git Support, which allows for version control. Visual Studio Code also includes robust debugging tools and supports multiple programming languages. Thus making it an ideal environment for iterative testing and continuous improvements.

5.0.3 Framework Used

The project was developed using Django as the primary web framework, with the latest version of Python of 3.12.4. Django was selected for its robustness and scalability, making it suitable to handle audio input and output. Python's extensive library integrates with Django seamlessly. This facilitates the efficient handling of audio data.

Django's Model-View-Template (MVT) architecture supports a workflow that is organized. It allows for modularization of processing audio. The framework has the ability to manage different data types and middle-ware streamline development especially when handling audio files.

Initially, Streamlit was considered as it has the capability to do rapid prototyping and its simplicity. It was valuable during the initial stages of development, especially visualizing and verifying the translated text in real time. However, Streamlit has its limitations when managing continuous input and output hinder the project's requirement for dynamic process-

ing. Consequently, the project was moved to Django where there was a greater amount of control over handling and data streaming. This change made the model's audio processing pipeline to be executed reliably.

5.0.4 Libraries Used

These are the libraries used and a brief explanation of how they play a role within development:

- `django.shortcuts`: The render function in Django was used in the managing how the user interacts with the web interface. This was important when handling the interface where audio inputs are submitted and received audio outputs.
- `gTTS`: This was the unofficial version. This library provided the text-to-speech conversion transforming the written outputs into audio responses. It facilitated the audio feedback sounding natural .
- `googletrans`: The Translator module was used to translate between languages. This was valuable in providing multilingual support for both input and output.
- `speech recognition`: this was used for the handling and converting audio input into text. This was crucial in feeding the speech received from the user, allowing for response and processing to be done in real time.
- `os` and `django.conf.settings`: These were used in the management of file paths and configuration. This made sure that audio files were saved correctly, accessed and processed within the Django framework.
- `base64` and `io.BytesIO`: These modules allow for the efficient decoding and encoding of audio data memory.
- `tempfile`: This allows for intermediate audio pre-processing steps to be carried out and securely without messing up the file system.
- `pydub`: This was used in the manipulation of audio files. This was converting formats, and segmenting and merging audio clips.
- `noisereducer`: The reduce noise function played a role in enhancing the clarity of the audio submitted by the user. It minimized the background, making sure that recognition was more accurate and reliable.

- numpy: it was used in applying math operations to data and handles audio waveforms.
- scipy: This was used for additional audio processing tasks.
- librosa: Was used in analyzing audio capabilities. This was used in the refinement of the audio analysis phase.

To rigorously test the effectiveness of the model, ten sentences -as displayed in tables 1 and 2- commonly used in classroom settings were selected. As Zulu is a Bantu language, special attention was paid to phonetic accuracy to make sure that the synthesized voice sounded as natural as possible and understandable in Zulu. Tests were conducted in various environments, including a small classroom [this classroom had no students as it was used during the Term 3 holidays] and a student residence room to test the impact of HRTF on the quality of the audio and localization. The prototype's ability to handle ambient noise, reverberation and spatial acoustics were determined by trying to simulating realistic acoustic conditions.

Table of Sentences Used In Testing

Table 1: Sentences translated from English to isiZulu

English	IsiZulu
My name is..	Igama lami ngingu..
I am 15 years old	Nginemiyaka yishumi nehlanu ubudala
Ma'am can you please explain	Nkosikazi, ngicela ungichazele
I do not think i understand	Angicabangi ukuthi ngiyaqonda
May I please go to the bathroom?	Ngicela ukuya endlini yangasese?
What will the test be on?	Yini ezobhalwa esivivinyweni?
Let's work in groups	Masisebenze ngamaqembu
Can you please repeat that	Ungakwazi ukuphinda lokho?
Please raise your hand if you do not understand	Ngicela uphakamise isandla uma ungaqondi
Remember to write you answer in english	Khumbula ukubhala impendulo yakho ngesiNgisi

Table 2: Sentences translated from English to Afrikaans

English	Afrikaans
My name is..	My naam is...
I am 15 years old	Ek is vyftien jaar oud
Ma'am can you please explain	Mevrou, kan u asseblief verduidelik?
I do not think i understand	Ek dink nie ek verstaan nie
May I please go to the bathroom?	Mag ek badkamer toe gaan asseblief?
What will the test be on?	Waarop sal die toets wees?
Let's work in groups	Laat ons in groepe werk
Can you please repeat that	Kan u dit herhaal asseblief?
Please raise your hand if you do not understand	Steek asseblief jou hand op as jy nie verstaan nie
Remember to write you answer in english	Onthou om jou antwoorde in Engels te skrfy

5.0.5 Results

In the beginning, Eurocentric languages -such as Afrikaans, French and Spanish- were used. When testing, they demonstrated strong performance. Audio outputs were accurate and natural compared to other applications supporting other languages. The efficacy can be attributed to the design of most audio processing and translation libraries, which focused predominantly on European languages. In comparison, African languages -namely isiZulu, Swahili, isiXhosa and Sotho- were tested which gave mixed results. Translations showcased some reasonably accurate accuracy which only reached approximately 60%.

Google translate was used for text translation for these languages. However an issue arose as it lacks vocal support for african languages. This brought down the utility in providing accurate audio output. Afrikaans performed adequately with gTTS as it shares structural similarity with European languages. At some point, Mozilla TTS was considered but was quickly disregarded. This was because of compatibility issues. It only supports Python versions 3.3 to 3.9 whereas the prototype was Python 3.12.4. Thus leading to constant switching between versions and in the end going with gTTs.

When it comes to handling audio, initial recordings were saved as blob files. This format made it difficult when it came to attempting to manipulate. Attempts to convert the blob files into MP3 files were done but ultimately gave more issues. All audio files were converted into WAV files, which made it more stable and processing was much smoother. Noise isolation was also performed in an attempt to get clear data.

HRTF was tried and tested to simulate spatial effects. Within a small student residence room and a small empty classroom, it performed fine. Many datasets were used to get the optimal parameters for HRTF. There were no perfect results -this because everyone's ears are different- and this could be looked into further. However, it is recommended that stereo earphones or headphones are to be used. This is as the audio experience is enhanced for spatial sound voice clarity

6 Challenges and Limitations of Model Application

There was a issue when it can to finding the optimal Head Related Transfer Function. It became too time-consuming which now underscores the need for automated and dynamic customization for tonal accuracy.

The isiZulu text translation were accurate to a certain degree, though many tonal nuances were often missed. As isiZulu is a tonal Bantu language, it require extreme attention to tone, which was not captured in the translation the prototype gave. Another issue was in regards to cultural context. The technologies utilized lack the awareness of these subtleties. For example, when greeting anyone who is older, the correct address would "Sanibonani" instead of "Sawubona". This is because you are not only greet the person themselves, but you are greeting their entire family and ancestors, even if they are not present [they are a somewhat representative]. As an attempt to having a natural voice synthesis, phonetics were used but had challenges tonal. Test-to-speech voices usually cater to Germanic and Latin rooted languages [41]. Using phonetics for simple phrases such as "Hello", "My Name is" displayed limited success. However, in complicated classroom settings, phonetic inaccuracies became more apparent especially, with the how Google Translate's synthesis has a problem with trying to produce the correct sounds.

From there, many TTS options were examined and were found to be unsuitable for South African languages. Most either lacked support for these languages or either were expensive [which would be unsustainable within an Honors project.].

- Amazon Polly: Intended for commercial use
- eSpeak: Supports some languages such as Swahili and Sestwana but not isiZulu
- Festival Speech Synthesis: Free but no South African languages
- Google Cloud Text-To-Speech: Requires payment for each word translated.

Thus, an unofficial version of Google Translate was used. This had its own issues as there was a risk of translation inaccuracies.

7 Conclusion and Future

This research focused on bridging the gap in SLEs to support multilingual communication. Thanks to DSR, a prototype was design and iteratively refined to incorporate real S2ST and perspective sharing. testing included phonetic improvements and adjustments to enhance accuracy. Whilst the prototype displayed effectiveness within European languages, constraints with TTS support for African languages and accuracy when it comes to tone needs to be further looked at. This research provided a foundational model that has the potential to be improved in the future.

Future works should emphasis on developing more accurate speech-to-speech translation systems for African languages. These systems would incorporate cultural context and tonal nuances to ensure authentic communication. This will involve the creation and training of synthetic voices specifically for these languages. They should aim to accurately reflect regional accents and speech characteristics. In addition further research into automatic Head Related Transfer Function processing within a SLE will need to be done. As this will allow for a seamless integration into not relying on fixed placeholder values. With this, a more adaptive and immersive experience would be had by dynamically adjusting the auditory cues to line up with the scenario at hand within a South African classroom.

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